



Mini Cooper, 1962

Mini cars

As design innovations continued, some cars—usually big, fast, gas-guzzlers—came to be seen as status symbols. Then in 1959, small and economical started to look good when designer Alec Issigonis launched the Mini. This car, the first of many models, had a space-saving transverse (sideways) mounted engine and fuel-saving front-wheel drive.

A plug-in electric car's charge port connects to an electricity supply to recharge batteries.



Going electric

In the 21st century, pollution caused by gas-burning cars is a serious problem. Major development of all-electric vehicles could mean cleaner air. These cars use rechargeable batteries to drive electric wheel motors. Most models are used over short distances but some, such as the Tesla Roadster, have a range of more than 185 miles (300 km) between recharges.



Headlights sat flush in curved pontoon fairings rather than sticking out and dragging on the passing air.

Road safety

With more than a billion vehicles on the roads worldwide, car designers and manufacturers take safety seriously.

► Crash test dummies

These lifelike human models are used to test the effects of crash impacts on people. This helps car engineers to work out how to reduce the risk of injuries.

► Seatbelts

Seatbelts, devices which save many lives each year, reduce the forward motion of a person's body if a vehicle stops suddenly.

► Airbag

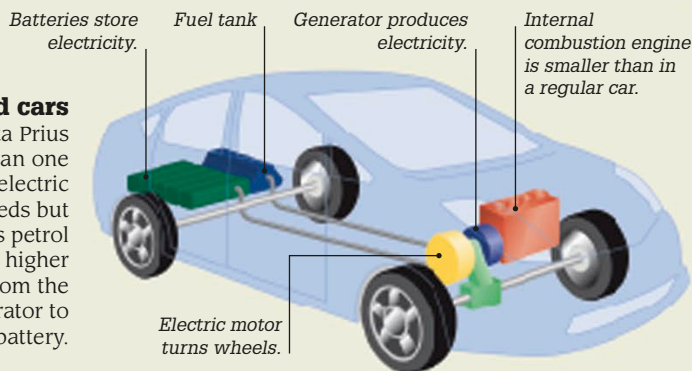
Sudden stops trigger airbags, which inflate in less than 0.05 seconds to cushion people inside a vehicle from harmful impact with the car interior.



Airbag and crash test dummy

Hybrid cars

Hybrid cars, such as the Toyota Prius launched in 1997, contain more than one form of propulsion. The Prius's electric motor moves the car at low speeds but works together with the car's petrol engine when acceleration and higher speeds are required. Energy from the car's movement drives the generator to recharge the electric motor's battery.



1933

Invented in the 1890s, diesel engines were first used in trucks and buses until the Citroën Rosalie became the first diesel-powered, mass-produced car.

1939

General Motors introduced the Hydra-Matic (a motor transmission that changed gears automatically) for their Cadillac and Oldsmobile ranges. It was the first mass-produced automatic transmission for passenger cars.

1973

The first catalytic converters for production cars were introduced. These convert toxic emissions from the engine into less harmful gases and water vapor.

1997

The Toyota Prius became the first mass-produced hybrid car, with battery-powered electric motors reducing fuel consumption by its gas engine.

Toyota Prius

